Time Series Analysis with Gaussian Processes

Workshop exercise session

Use the monthly Kericho dataset to explore temporal patterns and compare five Gaussian process covariance models.

Data and setup

Dataset: data/Kericho.csv. Variables include Year, Month, Cases, Rain, minimum and maximum temperature, and VCAP. Confirm the monthly ordering, inspect missing values, and report the observation period.

Exercise 1 Exploratory analysis and decomposition

1. Summarize Cases and Rain. Plot the monthly case series and describe trend, seasonal behavior, unusual values, and changing variance.
2. Apply STL decomposition with a periodic seasonal component. Interpret the trend, seasonal, and remainder panels. Is an additive decomposition reasonable?

Exercise 2 Dependence diagnostics

3. Plot ACF and PACF for the original series and detrended residuals. Which patterns could be explained by trend or annual seasonality?
4. Calculate the empirical temporal semivariogram for lags 1 to 48 months. Describe its rise, sill, and practical range.

Exercise 3 Fit five temporal models

Fit a constant-mean Gaussian process with measurement error using ML for each covariance: AR(1), Matérn v=0.5, Matérn v=1.5, and Matérn v=2.5.

5. State the covariance assumptions and report parameter estimates, log likelihood, convergence status, and AIC for each model.
6. Compute predictive RMSE, and AIC. Define the prediction design and interpret model rankings.

Exercise 4 Hold-out-validation

7. Create a training (80%) and test set (20%) and evaluate the performance of the model.